SMOOTHNESS AT A PRIME DETECTED BY SMALL LOCAL ARTINIAN LIFTING TESTS

THE STACKS PROJECT AUTHORS

Editorial note. Original author theorem and proof are retained. Only this target is counted; local supporting statements are not separate new entries.

Allowed standard prerequisites (editorial).

- Kahler differentials and localization, the Jacobian/split differential criterion for smoothness, Artin-Rees and split-injection descent from completion.. These are the named standard inputs; the selected main construction and its local nonroutine support remain in full.

Source reference. Original proposition 00TN: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1081–1081. Stable Stacks locator for declared standard input or surrounding context. No selected local proof is replaced by this pointer.

Source reference. Original lemma 00FP: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 579–579. Stable Stacks locator for declared standard input or surrounding context. No selected local proof is replaced by this pointer.

Source reference. Original section 00S0: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1022–1022. Stable Stacks locator for declared standard input or surrounding context. No selected local proof is replaced by this pointer.

Source reference. Original lemma 07BU: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9354–9354. Stable Stacks locator for declared standard input or surrounding context. No selected local proof is replaced by this pointer.

Source reference. Original lemma 02HQ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3273–3273. Stable Stacks locator for declared standard input or surrounding context. No selected local proof is replaced by this pointer.

Source reference. Original lemma 00RT: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1015–1015. Stable Stacks locator for declared standard input or surrounding context. No selected local proof is replaced by this pointer.

Source reference. Original lemma 00RU: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1016–1016. Stable Stacks locator for declared standard input or surrounding context. No selected local proof is replaced by this pointer.

Source reference. Original lemma 02HP: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3272–3272. Stable Stacks locator for declared standard input or surrounding context. No selected local proof is replaced by this pointer.

Date: stacks-project:algebra.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original lemma 00IN: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 685–685. Stable Stacks locator for declared standard input or surrounding context. No selected local proof is replaced by this pointer.

Source reference. Original lemma 02HO: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3271–3271. Stable Stacks locator for declared standard input or surrounding context. No selected local proof is replaced by this pointer.

Lemma 0.1. *Let $R \to S$ be a ring map. Let $\mathfrak{q}$ be a prime ideal of S lying over $\mathfrak{p} \subset R$. Assume R is Noetherian and $R \to S$ of finite type. The following are equivalent:*

1. *$R \to S$ is smooth at $\mathfrak{q}$,*
2. *for every surjection of local R-algebras $(B', \mathfrak{m}') \to (B, \mathfrak{m})$ with $\mathrm{Ker}(B' \to B)$ having square zero and every solid commutative diagram*

$$\begin{array}{ccc} S & \longrightarrow & B \\ \uparrow & \nearrow & \uparrow \\ R & \longrightarrow & B' \end{array}$$

   such that $\mathfrak{q} = S \cap \mathfrak{m}$ there exists a dotted arrow making the diagram commute,
3. *same as in (2) but with $B' \to B$ ranging over small extensions, and*
4. *same as in (2) but with $B' \to B$ ranging over small extensions such that in addition $S \to B$ induces an isomorphism $\kappa(\mathfrak{q}) \cong \kappa(\mathfrak{m})$.*

Proof. Assume (1). This means there exists a $g \in S$, $g \notin \mathfrak{q}$ such that $R \to S_g$ is smooth. By Proposition 00TN we know that $R \to S_g$ is formally smooth. Note that given any diagram as in (2) the map $S \to B$ factors automatically through $S_\mathfrak{q}$ and a fortiori through S_g. The formal smoothness of S_g over R gives us a morphism $S_g \to B'$ fitting into a similar diagram with S_g at the upper left corner. Composing with $S \to S_g$ gives the desired arrow. In other words, we have shown that (1) implies (2).

Clearly (2) implies (3) and (3) implies (4).

Assume (4). We are going to show that (1) holds, thereby finishing the proof of the lemma. Choose a presentation $S = R[x_1, \ldots, x_n]/(f_1, \ldots, f_m)$. This is possible as S is of finite type over R and therefore of finite presentation (see Lemma 00FP). Set $I = (f_1, \ldots, f_m)$. Consider the naive cotangent complex

$$\mathrm{d} : I/I^2 \longrightarrow \bigoplus\nolimits_{j=1}^{n} S\mathrm{d}x_j$$

of this presentation (see Section 00S0). It suffices to show that when we localize this complex at $\mathfrak{q}$ then the map becomes a split injection, see Lemma 07BU. Denote $S' = R[x_1, \ldots, x_n]/I^2$. By Lemma 02HQ we have

$$S \otimes_{S'} \Omega_{S'/R} = S \otimes_{R[x_1,\ldots,x_n]} \Omega_{R[x_1,\ldots,x_n]/R} = \bigoplus\nolimits_{j=1}^{m} S\mathrm{d}x_j.$$

Thus the map

$$\mathrm{d} : I/I^2 \longrightarrow S \otimes_{S'} \Omega_{S'/R}$$

is the same as the map in the naive cotangent complex above. In particular the truth of the assertion we are trying to prove depends only on the three rings $R \to S' \to S$. Let $\mathfrak{q}' \subset R[x_1, \ldots, x_n]$ be the prime ideal corresponding to $\mathfrak{q}$. Since localization commutes with taking modules of differentials (Lemma 00RT) we see that it suffices to show that the map

$$\mathrm{d} : I_{\mathfrak{q}'}/I_{\mathfrak{q}'}^2 \longrightarrow S_\mathfrak{q} \otimes_{S'_{\mathfrak{q}'}} \Omega_{S'_{\mathfrak{q}'}/R} \tag{0.1.1}$$

coming from $R \to S'_{\mathfrak{q}'} \to S_{\mathfrak{q}}$ is a split injection.

Let $N \in \mathbf{N}$ be an integer. Consider the ring

$$B'_N = S'_{\mathfrak{q}'}/(\mathfrak{q}')^N S'_{\mathfrak{q}'} = (S'/(\mathfrak{q}')^N S')_{\mathfrak{q}'}$$

and its quotient $B_N = B'_N/IB'_N$. Note that $B_N \cong S_{\mathfrak{q}}/\mathfrak{q}^N S_{\mathfrak{q}}$. Observe that B'_N is an Artinian local ring since it is the quotient of a local Noetherian ring by a power of its maximal ideal. Consider a filtration of the kernel I_N of $B'_N \to B_N$ by B'_N-submodules

$$0 \subset J_{N,1} \subset J_{N,2} \subset \ldots \subset J_{N,n(N)} = I_N$$

such that each successive quotient $J_{N,i}/J_{N,i-1}$ has length 1. (As B'_N is Artinian such a filtration exists.) This gives a sequence of small extensions

$$B'_N \to B'_N/J_{N,1} \to B'_N/J_{N,2} \to \ldots \to B'_N/J_{N,n(N)} = B'_N/I_N = B_N = S_{\mathfrak{q}}/\mathfrak{q}^N S_{\mathfrak{q}}$$

Applying condition (4) successively to these small extensions starting with the map $S \to B_N$ we see there exists a commutative diagram

$$\begin{array}{ccc} S & \longrightarrow & B_N \\ \uparrow & \searrow & \uparrow \\ R & \longrightarrow & B'_N \end{array}$$

Clearly the ring map $S \to B'_N$ factors as $S \to S_{\mathfrak{q}} \to B'_N$ where $S_{\mathfrak{q}} \to B'_N$ is a local homomorphism of local rings. Moreover, since the maximal ideal of B'_N to the Nth power is zero we conclude that $S_{\mathfrak{q}} \to B'_N$ factors through $S_{\mathfrak{q}}/(\mathfrak{q})^N S_{\mathfrak{q}} = B_N$. In other words we have shown that for all $N \in \mathbf{N}$ the surjection of R-algebras $B'_N \to B_N$ has a splitting.

Consider the presentation

$$I_N \to B_N \otimes_{B'_N} \Omega_{B'_N/R} \to \Omega_{B_N/R} \to 0$$

coming from the surjection $B'_N \to B_N$ with kernel I_N (see Lemma 00RU). By the above the R-algebra map $B'_N \to B_N$ has a right inverse. Hence by Lemma 02HP we see that the sequence above is split exact! Thus for every N the map

$$I_N \longrightarrow B_N \otimes_{B'_N} \Omega_{B'_N/R}$$

is a split injection. The rest of the proof is gotten by unwinding what this means exactly. Note that

$$I_N = I_{\mathfrak{q}'}/(I_{\mathfrak{q}'}^2 + (\mathfrak{q}')^N \cap I_{\mathfrak{q}'})$$

By Artin-Rees (Lemma 00IN) we find a $c \geq 0$ such that

$$S_{\mathfrak{q}}/\mathfrak{q}^{N-c} S_{\mathfrak{q}} \otimes_{S_{\mathfrak{q}}} I_N = S_{\mathfrak{q}}/\mathfrak{q}^{N-c} S_{\mathfrak{q}} \otimes_{S_{\mathfrak{q}}} I_{\mathfrak{q}'}/I_{\mathfrak{q}'}^2$$

for all $N \geq c$ (these tensor product are just a fancy way of dividing by $\mathfrak{q}^{N-c}$). We may of course assume $c \geq 1$. By Lemma 02HQ we see that

$$S'_{\mathfrak{q}'}/(\mathfrak{q}')^{N-c} S'_{\mathfrak{q}'} \otimes_{S'_{\mathfrak{q}'}} \Omega_{B'_N/R} = S'_{\mathfrak{q}'}/(\mathfrak{q}')^{N-c} S'_{\mathfrak{q}'} \otimes_{S'_{\mathfrak{q}'}} \Omega_{S'_{\mathfrak{q}'}/R}$$

we can further tensor this by $B_N = S_{\mathfrak{q}}/\mathfrak{q}^N$ to see that

$$S_{\mathfrak{q}}/\mathfrak{q}^{N-c} S_{\mathfrak{q}} \otimes_{S'_{\mathfrak{q}'}} \Omega_{B'_N/R} = S_{\mathfrak{q}}/\mathfrak{q}^{N-c} S_{\mathfrak{q}} \otimes_{S'_{\mathfrak{q}'}} \Omega_{S'_{\mathfrak{q}'}/R}.$$

Since a split injection remains a split injection after tensoring with anything we see that

$$S_{\mathfrak{q}}/\mathfrak{q}^{N-c} S_{\mathfrak{q}} \otimes_{S_{\mathfrak{q}}} (0.1.1) = S_{\mathfrak{q}}/\mathfrak{q}^{N-c} S_{\mathfrak{q}} \otimes_{S_{\mathfrak{q}}/\mathfrak{q}^N S_{\mathfrak{q}}} (I_N \longrightarrow B_N \otimes_{B'_N} \Omega_{B'_N/R})$$

is a split injection for all $N \geq c$. By Lemma 02HO we see that (0.1.1) is a split injection. This finishes the proof. □